Miguel Aenlle

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EDUCATION

Master of Science in Computer Science, University of Illinois Urbana-Champaign *August 2027 — May 2028*

Bachelor of Science in Computer Science, University of Illinois Urbana-Champaign *August 2024 — May 2027*

- **GPA:** 4.0/4.0
- **Relevant Coursework:** Machine Learning Systems, Distributed Systems, Cloud Computing, Operating Systems, Deep Learning

EXPERIENCE

Software Engineer — AI Infrastructure, **PrairieLearn** *August 2026 — Present*

- Created scalable coding agent platform automating STEM course development, autograder debugging, and SQL-based data analysis
- Engineered sandbox engine for untrusted AI-generated SQL/code, approval-gated commits, and stateless VM backup/restoration
- Deploying to 180,000-student platform using Terraform IaC, AWS, and Cloudflare with real-time cost accounting in PostgreSQL/Redis

ML Systems Research Intern, **UIUC SSAIL** *August 2026 — Present*

- Increased throughput of Qwen GPU serving cluster by 40% vs. baseline via LLM-driven autoresearch loop to tune scheduler policy
- Built a simulated GPU cluster environment where LLM agents write scheduler code, debug it, and iterate using benchmarks and traces

Software Engineer Intern, **Adobe** *May 2026 — August 2026*

- Led development of multi-agent system utilizing internal documents/presentations to generate export-ready marketing campaigns
- Implemented GCP-based transcription, search, and filtering pipeline for 10,000+ assets and Adobe Express templates per client
- Designed LLM-as-a-judge eval environment measuring output quality and latency across 30+ real enterprise marketing scenarios

AI/ML Engineer Intern, **PrairieLearn** *August 2025 — May 2026*

- Architected AI image grading pipeline achieving human-level 99% accuracy across 10,000+ handwritten mathematics submissions
- Enabled long-running, resumable multi-agent rubric editing by building PostgreSQL-backed durable workflow infrastructure on EC2
- Shipped to 180,000+ students via concurrent batch inference, socket-streamed grading progress, and usage-based Stripe billing

Software Engineer Intern, **PrairieLearn** *December 2024 — August 2025*

- Engineered image-capture API with 1ms capture processing for 180,000+ students in high-stakes exams via WebSockets and S3
- Redesigned staff navigation user experience across 80+ pages, streamlining instructor workflows for over 2,000 active courses

RESEARCH AND PROJECTS

ML Systems Engineer, **OptiTrain** *June 2026 — September 2026*

- Engineered distributed LLM training and inference platform with AWS GPU fleets, PyTorch DDP/NCCL, and failure recovery
- Chaos-tested 17-hour NanoGPT pretraining run on 8 GPU servers, self-healing 31 injected node failures with 167s median recovery
- Built high-throughput Go/Kubernetes inference engine, raising tokens per second via vLLM batching, KV caching, and dtype casting

Research Software Developer, **Google Summer of Code — Emory University** *June 2025 — February 2026*

- Trained specialized diffusion and GAN models to augment a limited, proprietary gastric MRI training dataset by 1,600% in size
- Raised UNet segmentation accuracy from 39% to 80% on held-out gastric scans via the expanded dataset and hyperparameter tuning

CONFERENCE PUBLICATIONS

Automated Grading of Handwritten Mathematics Using Vision-Capable LLMs *May 2026*

- Levine, J., **Aenlle, M.**, Zilles, C., West, M., Silva, M. *International Conference on AI in Education (AIED)*, 2026. arXiv:2605.19043

LEADERSHIP AND ACTIVITIES

Web Team Lead, **HackIllinois** *July 2025 — March 2026*

- Led 4-engineer team to ship admin, logistics, and marketing systems for UIUC's flagship hackathon, garnering 500,000+ page views
- Shipped registration and logistics platform serving 2,000+ attendees and 50+ staff members across three-day flagship event

TECHNICAL SKILLS

Languages: Python, Go, C++, C, Java, TypeScript, JavaScript, SQL, Bash, Swift, HTML, CSS

Systems / Cloud: GCP (Vertex AI, Compute), AWS (EC2, S3, Lambda), Cloudflare (Workers, R2), Kubernetes, Docker, Linux, Prometheus

AI / ML: PyTorch, torch.distributed, DDP, NCCL, NVLink, CUDA, vLLM, LangGraph, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas

Web / Data: React, Node.js, Express, REST, tRPC, WebSockets, SocketIO, Stripe API, PostgreSQL, Redis, MongoDB, CI/CD, Git